AI Engineer Curriculum — v6

Rajesh Ramani · 14 Projects · 4-Month Self-Training Program · Updated 2026-04-29

Progress

P1–P6 COMPLETE · 6 of 13 projects (46%)

Quiz scores P1: 7/7 P2: 6.5/8 P3: 6.5/7 P4: 6.5/8 P5: 6/6 P6: 5/7

v6 Changes (2026-04-29) — Driven by 8 Real Job Postings

Gap analysis run against Oracle Observability Principal, Oracle Health Senior + Principal AI Agent, Salesforce Agent Systems, Adobe Sr AI Systems, Airwallex AI Engineer, Creao AI Agent Engineer, Synopsys Sr AI Engineer.

- **P7 expanded** — added Ragas, DeepEval, Promptfoo + regression suite (`#79`)
- **P10 expanded** — added FastAPI, Pinecone, multi-cloud LLM routing (Bedrock + Azure + Vertex), reliability patterns, cost tracking (`#74–#78`)
- **P12 NEW** — AutoGen vs LangGraph vs CrewAI framework showdown + LinkedIn blog (`#80`)

Project Roadmap

#	Project	Folder	Key Tech	Tickets	Status
P1	ToolBot — CLI agent, 3 LLMs	p1_toolbot	async, Pydantic v2, tool_use, streaming, multi-LLM	#17–20	DONE 7/7
P2	DocTalk — PDF Q&A + citations	p2_doctalk	Embeddings, Chroma → Qdrant, RAG, LangChain	#21–24	DONE 6.5/8

#	Project	Folder	Key Tech	Tickets	Status
P3	ResearchBot — multi-step web research	p3_researchbot	LangGraph StateGraph, checkpointing, HITL, Kafka, A2A intro	#25–28, #65, #66	DONE 6.5/7
P3-Adv	Advanced LangGraph — Send API, Orchestrator-Worker	p3_adv_langgraph	Send API, Flow Engineering, Temporal+LangGraph	#72	BACKLOG
P4	StockSage — stock analysis agent	p4_stocksage	Multi-tool, domain RAG, Mem0, structured output	#29–33	DONE 6.5/8
P5	ReviewCrew — GitHub PR reviewer	p5_reviewcrew	CrewAI, parallel execution, BaseTool, hierarchical	#34–36, #71	DONE 6/6
P6	MCP++ — StockSage MCP server + agentic client	p6_mcp	MCP spec, FastMCP, streamable-http, OAuth 2.1, A2A	#37–39, #73	DONE 5/7
—	**#69 Python Fundamentals**	p0_fundamentals	17 sections: lists, dicts, lambdas, decorators, async, ABC, Annotated, pathlib...	#69	NEXT
P7	Customer Health Monitor — observability + **evals**	p7_observable	LangSmith, LangFuse, Faker, churn detection, Ragas DeepEval Promptfoo	#40–42, **#79**	QUEUED
P8	PII Compliance Agent — agent security	p8_security	OWASP LLM Top 10, PII detection, NeMo Guardrails, audit trails	#48	BACKLOG
P9	Account Memory Agent — long-term memory	p9_memory	Zep (Graphiti temporal KG), Mem0, LongMemEval benchmark	#49	BACKLOG

#	Project	Folder	Key Tech	Tickets	Status
P10	AgentPlatform — production deploy EXPANDED v6	p10_platform	Docker, k3s, GitHub Actions, AWS Bedrock, Terraform FastAPI + SSE Pinecone Multi-cloud routing Reliability (tenacity, pybreaker) Cost tracking	#43–45, **#74–78**	BACKLOG
P11	Contract & Invoice Analyzer — vision	p11_multimodal	PyMuPDF, Claude vision, base64 image blocks, multi-modal RAG	#50	BACKLOG
P12	Multi-Agent Framework Showdown NEW v6	p12_framework_showdown	Same task built in AutoGen + LangGraph + CrewAI; LoC, latency, cost comparison; LinkedIn blog	**#80**	NEW

Concepts Mastered (P1–P6)

Project	Concepts
P1	async / await, Pydantic v2 (BaseModel, field_validator, model_dump), tool_use loop, streaming responses, system prompts, context truncation, multi-turn memory
P2	embeddings (sentence-transformers), cosine similarity, chunking strategies, RAG pipeline (ingest → embed → store → retrieve → answer), citation tracking, LangChain abstractions, Chroma → Qdrant
P3	LangGraph StateGraph, typed state, nodes/edges, conditional routing, MemorySaver / SqliteSaver checkpointing, Human-in-the-Loop (interrupt + update_state), `Annotated[list, operator.add]` reducer pattern, Kafka event streaming
P4	multi-tool orchestration, structured output, domain RAG, agent synthesis, financial APIs (yfinance), persistent ChromaDB

Project	Concepts
P5	CrewAI agents/tasks/crew, agent specialization (role/goal/backstory), parallel execution (Process.parallel), BaseTool subclassing, Pydantic args_schema, hierarchical agents, post-crew Python dispatch
P6	MCP spec (3 primitives — Tools, Resources, Prompts), FastMCP, streamable-http transport (vs deprecated SSE), tool schemas via inputSchema, resource templates ({param}), prompt template double-wrapping bug, generic tool dispatch (`session.call_tool(name, input)`), async with context managers, asyncio.gather concurrent tool calls

Concepts Coming Up (P7–P12)

Project	Concepts
P7	LangSmith tracing, span hierarchy, evaluator functions, LangFuse self-hosted dashboards, Faker synthetic data, **Ragas (context precision, faithfulness, answer relevancy), DeepEval (LLM unit tests), Promptfoo (A/B prompt testing), regression suite that fails PR on quality drop**
P8	OWASP LLM Top 10, PII detection (Presidio), prompt injection blocking, NeMo Guardrails, audit trails
P9	Zep temporal knowledge graphs (Graphiti), Mem0 semantic facts, LongMemEval benchmark comparison (63.8% vs 49%)
P10	Docker multi-stage builds, k8s manifests, secrets management, GitHub Actions CI/CD, AWS Bedrock, Terraform IaC, **FastAPI + SSE streaming endpoints, Pinecone (cloud vector DB), multi-cloud LLM routing (Bedrock + Azure OpenAI + GCP Vertex), tenacity exponential backoff, pybreaker circuit breaker, automatic provider fallback (Claude 429 → GPT-4o), per-request token + USD cost tracking with budget alerts**
P11	PyMuPDF page rendering as images, Claude vision API, base64 image content blocks, multi-modal RAG
P12	AutoGen conversational multi-agent pattern, framework comparison methodology (LoC, latency, cost, debuggability), technical writing for LinkedIn

Job-Posting Drivers for v6

Posting	Curriculum addition triggered
Oracle Principal SE — Agentic AI (Observability)	FastAPI, multi-cloud routing, reliability patterns
Oracle Health Senior + Principal AI Agent	Eval frameworks, governance, model selection
Salesforce AI Engineer — Agent Systems	Eval, observability, cost optimization, AutoGen
Adobe Sr AI Systems — Agentic + Productivity	Evaluation pipelines, multi-model routing
Airwallex AI Engineer	Framework comparison, RAG variations, Pinecone
Creao AI Agent Engineer	Vector DB choices (Pinecone), evaluation methods
Synopsys Sr AI Engineer — Enterprise Agentic	MCP + A2A (already in P6)

Supplementary Tickets (non-blocking)

- `#67` Raw ReAct Loop (react_agent.py)
- `#68` Reflection / Self-Critique Pattern
- `#69` **Python Fundamentals for AI Engineering** — DOING NEXT (17 sections in p0_fundamentals/)
- `#71` P5 ReviewCrew Hierarchical process
- `#72` P3-Adv LangGraph Send API
- `#73` P6 OAuth 2.1
- `#74–78` P10 expansions (FastAPI, Pinecone, multi-cloud, reliability, cost)
- `#79` P7 eval frameworks deep dive
- `#80` P12 framework showdown

Multi-LLM Coverage

All three major LLMs are used across the curriculum — not Claude-only:

- **Anthropic Claude** (claude-sonnet-4-6 / claude-opus-4-7) — primary, tool_use, MCP, vision
- **OpenAI** (gpt-4o) — function calling, multi-modal comparison
- **Google Gemini** — function calling, multi-modal